Asymptotic Probabilities for Formulas of Propositional Intuitionistic Logic

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Abstract

We prove a zero-one law for model validity in intuitionistic propositional logic: for every formula φ, the proportion of n-element Kripke models that validate φ tends to 0 or to 1 as n tends to infinity. In contrast with the modal case, where every frame carries the same number of models, the requirement that intuitionistic valuations be up-sets makes the uniform measure on models a non-uniform measure on frames, so that the Kleitman–Rothschild theorem on the structure of almost all finite partial orders cannot be applied directly. We overcome this by encoding up-sets as additional minimal points, which realises the measure on models as the uniform measure on partial orders conditioned on an event whose probability is bounded away from zero. We further axiomatise the set of almost surely valid formulae: for each k there is a formula σ_k such that a formula φ in the variables $p_1, \ldots, p_k$ is almost surely valid if and only if $\sigma_k \to \varphi$ is intuitionistically derivable.

1 Introduction

Background and motivation. Ever since the work of Erdős [8], a certain question is often explored for a given property P of finite structures, namely: what is the probability that a randomly selected structure, chosen among finite structures of the same size, under a suitable probability measure, satisfies P? Even more intriguingly, what are the asymptotic probabilities? Specifically, do these probabilities converge to a limit as the size of the domain grows to infinity? In some cases, when the properties in question can be expressed by formulae in a particular logic, a "zero-one law" may apply: the probabilities converge to either 0 or 1. Thus, we can say that the formula is *almost surely false* or *almost surely true*.

The systematic study of logical zero-one laws began independently in the works of Glebskiĭ et al. [20] and Fagin [9]. They established a zero-one law for first-order classical logic over purely relational vocabularies. The approach in [20] involves a complex quantifier elimination method, while Fagin [9] employs the so-called extension axioms, which form a complete theory characterizing a countable random structure. Lynch [16] attributes the discovery of this theory to Stanisław Jaśkowski, although both the theory and the associated random structure were first published in [10], where Gaifman noted it as an example proposed by Michael Rabin and Dana Scott. For further insights on the classical story, readers may consult [7, 13] for accessible introductions to zero-one laws and [6] for a comprehensive survey.

More relevantly to the present paper, zero-one laws have also been studied for modal logics. Halpern and Kapron [14] proved zero-one laws for model validity in the modal logics K, T, $\mathsf{S4}$ and $\mathsf{S5}$, and proposed

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corresponding laws for frame validity. Le Bars [1] subsequently showed that the zero-one law fails for frame satisfiability in K. More recently Verbrugge [19] proved zero-one laws for the provability logic GL with respect to both models and frames, axiomatised the formulae valid in almost all models and in almost all frames, and showed that the axiomatisation of almost sure frame validity for S4 given in [14] is not correct. The key combinatorial ingredient in [19], as in the present paper, is the theorem of Kleitman and Rothschild [15] describing the structure of almost all finite partial orders.

Since intuitionistic propositional logic has a Kripke semantics closely related to that of S4, it is natural to ask for a zero-one law in this setting. The question is not answered by the modal results. In each of the modal settings above a valuation assigns arbitrary sets of points to the variables, so that every frame on n points carries the same number of models, and a property of almost all frames is automatically a property of almost all models. An intuitionistic valuation must assign up-sets, and the number of up-sets depends on the frame. Consequently a uniformly random intuitionistic model does not have a uniformly random frame, and the Kleitman–Rothschild theorem does not apply to it directly; nor, as we show in Section 2.5, can the intuitionistic law be obtained from the law for S4 through the Gödel translation.

Contributions. We prove a zero-one law for model validity in intuitionistic propositional logic (Theorem 5.2): for each formula φ, the proportion of n-element models in which φ is valid tends to 0 or to 1. The difficulty just described is overcome by an elementary encoding (Lemma 3.5): an up-set can be represented by an additional minimal point lying below exactly its elements, and in this way the n-element models over k variables correspond bijectively to the partial orders on $n+k$ points in which k designated points are minimal. These form a proportion of all partial orders on $n+k$ points that tends to 4^{-k} (Lemma 3.7), so every property of almost all partial orders transfers to almost all models (Corollary 3.9). It follows that almost every model is bisimilar to a single finite model $\mathcal{A}_k$ (Theorem 4.4), and a formula is almost surely valid precisely when it is valid in $\mathcal{A}_k$. As a by-product we obtain the asymptotic number of models (Corollary 3.8). Finally, we axiomatise the almost surely valid formulae (Theorem 6.21): for each k we exhibit a formula σ_k such that a formula φ in the variables $p_1, \dots, p_k$ is almost surely valid if and only if $\sigma_k \to \varphi$ is intuitionistically derivable.

Organisation. Section 2 fixes notation and explains why the Gödel translation does not yield the result. Section 3 contains the encoding and the structure of almost all models. Sections 4 and 5 prove the zero-one law, Section 6 gives the axiomatisation, and Section 7 lists open problems.

2 Preliminaries

2.1 Syntax and Kripke semantics

Throughout the paper we fix an integer $k \geq 1$, write $K = \{1, \dots, k\}$, and work with the propositional variables $p_1, \dots, p_k$. The set $\mathcal{L}_k$ of formulae of intuitionistic propositional logic over these variables is generated by

$$\varphi \;::=\; p_i \mid \bot \mid (\varphi \wedge \varphi) \mid (\varphi \vee \varphi) \mid (\varphi \to \varphi), \qquad i \in K,$$

with $\neg\varphi$ abbreviating $\varphi \to \bot$, $\top$ abbreviating $\neg\bot$, and $\varphi \leftrightarrow \psi$ abbreviating $(\varphi \to \psi) \wedge (\psi \to \varphi)$. We write $\vdash_{\mathsf{IPC}} \varphi$ for derivability in intuitionistic propositional logic.

Let $(W, \leq)$ be a poset. For $w \in W$ we write $\uparrow w = \{v \in W : w \leq v\}$, and we call $U \subseteq W$ an *up-set* if $\uparrow w \subseteq U$ whenever $w \in U$. A subset U is an up-set precisely when it is a union of sets of the form $\uparrow w$, and the up-sets of $(W, \leq)$ form a Heyting algebra under $\subseteq$. We say that v *covers* w, and that w is a *subcover* of v, in symbols $w \prec v$, if $w < v$ and there is no u with $w < u < v$.

Definition 2.1. *A* model *is a triple $\mathfrak{M} = (W, \leq, V)$ in which $(W, \leq)$ is a finite poset and V assigns to each variable p_i an up-set $V(p_i) \subseteq W$. The poset $(W, \leq)$ is the* frame *of $\mathfrak{M}$.*

The requirement that each $V(p_i)$ be an up-set is the persistence condition. It has no counterpart in the modal settings discussed in the introduction, where valuations are arbitrary, and it is, as Section 3 explains, the source of the main difficulty. For a point $w \in W$ we write

$$V(w) \;=\; \{\, i \in K : w \in V(p_i) \,\}$$

for the set of indices of the variables that hold at w; thus $V(w) \subseteq V(v)$ whenever $w \leq v$. Since V is applied to variables in the one case and to points in the other, no confusion will arise.

Truth is defined at a point by the usual clauses:

$$\begin{aligned}
\mathfrak{M}, w \models p_i &\iff w \in V(p_i);\\
\mathfrak{M}, w \not\models \bot;&\\
\mathfrak{M}, w \models \varphi \wedge \psi &\iff \mathfrak{M}, w \models \varphi \text{ and } \mathfrak{M}, w \models \psi;\\
\mathfrak{M}, w \models \varphi \vee \psi &\iff \mathfrak{M}, w \models \varphi \text{ or } \mathfrak{M}, w \models \psi;\\
\mathfrak{M}, w \models \varphi \to \psi &\iff \text{for all } v \geq w,\ \mathfrak{M}, v \models \varphi \text{ implies } \mathfrak{M}, v \models \psi.
\end{aligned}$$

Lemma 2.2 (Persistence)**.** *For every formula φ and all $w \leq v$ in $\mathfrak{M}$, if $\mathfrak{M}, w \models \varphi$ then $\mathfrak{M}, v \models \varphi$.*

Proof. Induction on φ. The base case is the assumption that each $V(p_i)$ is an up-set, and the implication case uses transitivity of $\leq$. □

Definition 2.3. *A formula φ is* valid in the model $\mathfrak{M}$*, written $\mathfrak{M} \models \varphi$, if $\mathfrak{M}, w \models \varphi$ for every $w \in W$. It is* valid in the frame $(W, \leq)$ *if $(W, \leq, V) \models \varphi$ for every valuation V.*

By Lemma 2.2, $\mathfrak{M} \models \varphi$ if and only if $\mathfrak{M}, w \models \varphi$ at every minimal $w \in W$; in particular, if $\mathfrak{M}$ is *rooted*, that is, has a least element, its *root*, then $\mathfrak{M} \models \varphi$ if and only if φ is true at the root. A *generated submodel* of $\mathfrak{M}$ is the restriction of $\mathfrak{M}$ to an up-set of W; truth of every formula at every point of a generated submodel agrees with its truth in $\mathfrak{M}$.

This paper concerns model validity throughout. Frame validity is a substantially different question, which we leave open; see Section 7.

2.2 Bisimulations

Definition 2.4. *Let $\mathfrak{M} = (W, \leq, V)$ and $\mathfrak{N} = (W', \leq', V')$ be models. A* bisimulation *between $\mathfrak{M}$ and $\mathfrak{N}$ is a relation $Z \subseteq W \times W'$ such that whenever wZw':*

$$\begin{aligned}
&w \in V(p_i) \text{ if and only if } w' \in V'(p_i), \text{ for each } i \in K && \text{(atoms)}\\
&\text{if } w \leq v \text{ then there is } v' \in W' \text{ with } w' \leq' v' \text{ and } vZv' && \text{(forth)}\\
&\text{if } w' \leq' v' \text{ then there is } v \in W \text{ with } w \leq v \text{ and } vZv' && \text{(back)}
\end{aligned}$$

We call Z total *if every $w \in W$ satisfies wZw' for some w', and* surjective *if every $w' \in W'$ satisfies wZw' for some w. Points w and w' are* bisimilar *if wZw' for some bisimulation Z, and $\mathfrak{M}$ and $\mathfrak{N}$ are* bisimilar *if there is a bisimulation between them that is both total and surjective. See [12] for background.*

The totality and surjectivity requirements are essential: the empty relation satisfies (atoms), (forth) and (back) vacuously, so without them every two models would count as bisimilar. They are exactly what is needed for Corollary 2.6 below, which is how bisimulations are used in Sections 4 and 6.

Theorem 2.5. *Let Z be a bisimulation between $\mathfrak{M}$ and $\mathfrak{N}$ and suppose wZw'. Then for every formula $\varphi \in \mathcal{L}_k$,*

$$\mathfrak{M}, w \models \varphi \quad \textit{if and only if} \quad \mathfrak{N}, w' \models \varphi.$$

Proof. Induction on φ. The atomic case is (atoms) and the case of $\bot$ is trivial; conjunction and disjunction are immediate. For $\varphi = \psi \to \chi$, suppose $\mathfrak{M}, w \models \psi \to \chi$ and let $v' \geq' w'$ with $\mathfrak{N}, v' \models \psi$. By (back) there is $v \geq w$ with vZv', so $\mathfrak{M}, v \models \psi$ by the induction hypothesis, hence $\mathfrak{M}, v \models \chi$, hence $\mathfrak{N}, v' \models \chi$ again by the induction hypothesis. The converse direction uses (forth) symmetrically. □

Corollary 2.6. *If $\mathfrak{M}$ and $\mathfrak{N}$ are bisimilar then $\mathfrak{M} \models \varphi$ if and only if $\mathfrak{N} \models \varphi$, for every $\varphi \in \mathcal{L}_k$.*

Proof. Let Z be a total surjective bisimulation and suppose $\mathfrak{N} \not\models \varphi$, say $\mathfrak{N}, w' \not\models \varphi$. By surjectivity there is w with wZw', and then $\mathfrak{M}, w \not\models \varphi$ by Theorem 2.5. Totality gives the converse. □

Definition 2.7. *A* p-morphism *from $\mathfrak{M} = (W, \leq, V)$ to $\mathfrak{N} = (W', \leq', V')$ is a map $f \colon W \to W'$ such that, for all $w, v \in W$ and $v' \in W'$,*

(i) $w \leq v$ *implies* $f(w) \leq' f(v)$;

(ii) $f(w) \leq' v'$ *implies* $v' = f(v)$ *for some* $v \geq w$;

(iii) $w \in V(p_i)$ *if and only if* $f(w) \in V'(p_i)$, *for each* $i \in K$.

The graph of a p-morphism is a total bisimulation: (iii) is (atoms), (i) gives (forth) and (ii) gives (back). Hence, if there is a surjective p-morphism from $\mathfrak{M}$ onto $\mathfrak{N}$, then $\mathfrak{M}$ and $\mathfrak{N}$ are bisimilar.

2.3 Asymptotic probabilities

Definition 2.8. *For $n \geq 1$ let $\mathrm{Mod}_n^{\mathsf{IPC}}$ denote the set of models whose underlying set is $\{1, \ldots, n\}$, and let $\mathrm{Mod}_n^{\mathsf{IPC}}(\varphi)$ denote the subset of those in which φ is valid. We equip $\mathrm{Mod}_n^{\mathsf{IPC}}$ with the uniform probability measure.*

Recall that k, the number of propositional variables, is fixed. By Definition 2.1, a model with underlying set $\{1, \ldots, n\}$ consists of a poset F on $\{1, \ldots, n\}$ together with an independent choice of up-set of F for each of the k variables. Writing $U(F)$ for the number of up-sets of F, there are therefore exactly $U(F)^k$ models with frame F, and

$$|\,\mathrm{Mod}_n^{\mathsf{IPC}}\,| \;=\; \sum_F U(F)^k, \tag{1}$$

the sum ranging over all posets on $\{1, \ldots, n\}$. The factor $U(F)$ depends on F, and it is this dependence that makes the uniform measure on models a non-uniform measure on frames. Section 3 is where that is dealt with.

Definition 2.9. *A formula $\varphi \in \mathcal{L}_k$ is* almost surely valid *if*

$$\lim_{n\to\infty} \frac{|\,\mathrm{Mod}_n^{\mathsf{IPC}}(\varphi)|}{|\,\mathrm{Mod}_n^{\mathsf{IPC}}\,|} \;=\; 1,$$

and almost surely invalid *if the same limit is* 0.

Almost sure invalidity says that the proportion of models validating φ tends to zero. It does not say that $\neg\varphi$ is almost surely valid, and in general it does not imply this: p_1 and $\neg p_1$ are both almost surely invalid. The content of the zero-one law is that no formula falls between these two cases, that is, that the limit above always exists and lies in $\{0,1\}$. This is proved as Theorem 5.2; the set of almost surely valid formulae is axiomatised in Theorem 6.21.

Remark 2.10. *The measure depends on k, and k is fixed. Since every formula contains finitely many variables this suffices for the zero-one law as stated, and we show in Remark 5.4 that whether a given formula is almost surely valid does not depend on the choice of k. The encoding of Section 3 adjoins one point per variable and so requires k to be finite; we return to this in Section 7.*

2.4 An example: one variable

Example 2.11. *We anticipate the main results in order to illustrate them in the simplest case. Take $k = 1$ and write p for p_1. The representative model $\mathcal{A}_1$ constructed in Section 4 has a bottom point $\bot$, two middle points $m_\emptyset$ and $m_{\{1\}}$, at which p is false and true respectively, and a top point $\top$, at which p is true. Let $\mathbf{C}_2$ be the two-element chain*

$$\mathbf{C}_2 \;=\; (\{\bot,\top\},\ \bot<\top,\ V(p)=\{\top\}).$$

The relation $\{(\bot,\bot),(m_\emptyset,\bot),(m_{\{1\}},\top),(\top,\top)\}$ is a total surjective bisimulation between $\mathcal{A}_1$ and $\mathbf{C}_2$. So, by Theorem 5.2 and Corollary 2.6, a one-variable formula is almost surely valid if and only if it is valid in $\mathbf{C}_2$, and by Lemma 2.2 this holds if and only if it is true at $\bot$.

The up-sets of $\mathbf{C}_2$ form the three-element Heyting chain $\mathbf{3} = \{0<a<1\}$, with $V(p)=\{\top\}$ the middle element a. Evaluating at $\bot$ therefore amounts to computing the value of φ in $\mathbf{3}$ under $p\mapsto a$, and φ is almost surely valid exactly when that value is 1. For instance

$$\neg\neg p\mapsto 1,\qquad \neg p\vee\neg\neg p\mapsto 1,\qquad p\vee\neg p\mapsto a,\qquad \neg\neg p\to p\mapsto a,\qquad \neg p\mapsto 0,$$

so $\neg\neg p$ and weak excluded middle are almost surely valid, while excluded middle and double negation elimination are almost surely invalid. Although there are infinitely many pairwise non-equivalent one-variable formulae, by the Rieger–Nishimura theorem [17, 18], they take only three values in $\mathbf{3}$. If the value of φ is 1 then φ is almost surely valid; if it is 0 then φ is almost surely invalid but $\neg\varphi$, whose value is $\neg 0 = 1$, is almost surely valid; and if it is a then φ and $\neg\varphi$ are both almost surely invalid.

This agrees with Theorem 6.21: for $k=1$ the axiom σ_1 is equivalent over IPC to $\neg\neg p$ (Remark 6.4), so a one-variable formula is almost surely valid if and only if $\vdash_{\mathsf{IPC}} \neg\neg p\to\varphi$.

2.5 The Gödel translation does not give the result

Since IPC embeds into $\mathsf{S4}$ and a zero-one law for model validity in $\mathsf{S4}$ is already known [14], it is natural to hope that the intuitionistic law follows by translation. It does not, and it is worth saying precisely why before we begin.

A model for $\mathsf{S4}$ is a triple (W,R,V) with R a preorder on W and V assigning to each variable an arbitrary subset of W, modal formulae being interpreted in the usual way. We write $\mathrm{Mod}_n^{\mathsf{S4}}$ for the set of such models on $\{1,\dots,n\}$, again with the uniform measure, and $\mathrm{Mod}_n^{\mathsf{S4}}(\psi)$ for the set of those in which the modal formula ψ is valid. The Gödel translation [11] $T\colon \mathcal{L}_k\to\mathcal{L}_k^{\Box}$ is defined by

$$T(p_i)=\Box p_i,\qquad T(\bot)=\bot,\qquad T(\varphi\to\psi)=\Box\big(T(\varphi)\to T(\psi)\big),$$
$$T(\varphi\wedge\psi)=T(\varphi)\wedge T(\psi),\qquad T(\varphi\vee\psi)=T(\varphi)\vee T(\psi).$$

Definition 2.12. *Let $\mathfrak{N} = (W, R, V)$ be a model for* S4*. Write $w \sim v$ if wRv and vRw, so that $\sim$ is an equivalence relation on W, and let $\leq$ be the partial order induced by R on $W/{\sim}$. Put*

$$\mathrm{int}[\mathfrak{N}] \;=\; \big(W/{\sim},\, \leq,\, \widetilde{V}\big), \qquad \widetilde{V}(p_i) \;=\; \big\{[w] : vRu \text{ implies } u \in V(p_i), \text{ for all } v \in [w] \text{ and } u \in W\big\}.$$

Each $\widetilde{V}(p_i)$ is well defined on classes and is an up-set of $(W/{\sim}, \leq)$, by transitivity of R, so $\mathrm{int}[\mathfrak{N}]$ is a model in the sense of Definition 2.1. The adequacy of the translation is standard; see for instance [5, Lemma 3.81].

Proposition 2.13. *Let $\mathfrak{N}$ be a model for* S4*, let w be a point of $\mathfrak{N}$ and let $\varphi \in \mathcal{L}_k$. Then*

$$\mathrm{int}[\mathfrak{N}], [w] \models \varphi \quad \text{if and only if} \quad \mathfrak{N}, w \models T(\varphi).$$

The zero-one law for S4 applies in particular to the formulae $T(\varphi)$: for every $\varphi \in \mathcal{L}_k$,

$$\lim_{n\to\infty} \frac{|\,\mathrm{Mod}_n^{\mathsf{S4}}(T(\varphi))|}{|\,\mathrm{Mod}_n^{\mathsf{S4}}\,|} \;\in\; \{0, 1\},$$

and by Proposition 2.13 this is the limiting proportion of S4-models $\mathfrak{N}$ for which φ is valid in $\mathrm{int}[\mathfrak{N}]$. This is a statement about the uniform measure on $\mathrm{Mod}_n^{\mathsf{S4}}$. To turn it into a zero-one law for IPC one would need to know that φ is valid in almost all IPC-models if and only if $T(\varphi)$ is valid in almost all S4-models. That is false.

Proposition 2.14. *The formula $\neg\neg p_1$ is almost surely valid, whereas for every n at most half of the models in $\mathrm{Mod}_n^{\mathsf{S4}}$ validate $T(\neg\neg p_1)$.*

Proof. In any finite model, $\neg\neg p_1$ holds at a point w if and only if every $v \geq w$ lies below some point at which p_1 holds; so $\neg\neg p_1$ is valid in a finite model if and only if p_1 holds at each of its maximal points. In particular $\neg\neg p_1$ is valid in the model $\mathcal{A}_k$ of Definition 4.1, and is therefore almost surely valid by Theorem 5.2.

Now let $\mathfrak{N} = (W, R, V)$ be a model for S4. By Proposition 2.13, $T(\neg\neg p_1)$ is valid in $\mathfrak{N}$ if and only if $\neg\neg p_1$ is valid in $\mathrm{int}[\mathfrak{N}]$, that is, if and only if every maximal element $[z]$ of $(W/{\sim}, \leq)$ belongs to $\widetilde{V}(p_1)$. For such a maximal $[z]$, the conditions $v \in [z]$ and vRu force $u \in [z]$, so $[z] \in \widetilde{V}(p_1)$ exactly when $[z] \subseteq V(p_1)$. Fix a preorder R on $\{1, \ldots, n\}$ and a point z whose class is maximal. Exactly half of the valuations V have $z \in V(p_1)$, so at most half of the models with underlying preorder R validate $T(\neg\neg p_1)$. Summing over R gives the claim. □

The underlying reason is that the two measures live on very different sets of models.

Proposition 2.15. *For every $k \geq 1$,*

$$\frac{|\,\mathrm{Mod}_n^{\mathsf{IPC}}\,|}{|\,\mathrm{Mod}_n^{\mathsf{S4}}\,|} \;\leq\; 2^{-kn/2 + O_k(\log n)} \;\longrightarrow\; 0.$$

Proof. Let P_m denote the number of posets on an m-element set. Every poset on $\{1, \ldots, n\}$ is a preorder and every function from $\{p_1, \ldots, p_k\}$ to subsets of $\{1, \ldots, n\}$ is an S4-valuation, so $|\,\mathrm{Mod}_n^{\mathsf{S4}}\,| \geq P_n 2^{kn}$. On the other hand $|\,\mathrm{Mod}_n^{\mathsf{IPC}}\,| \leq P_{n+k}$ by Lemma 3.5 below, and $P_{n+k} = P_n 2^{kn/2 + O_k(\log n)}$ by the estimate of P_m in Theorem 3.1. □

So $\mathrm{Mod}_n^{\mathsf{IPC}}$ is a null subclass of $\mathrm{Mod}_n^{\mathsf{S4}}$, and knowing how almost all S4-models behave places no constraint on how almost all IPC-models behave. The intuitionistic zero-one law must be proved directly. The same phenomenon explains why the proof cannot simply invoke the Kleitman–Rothschild theorem: sampling an IPC-model uniformly does not sample its frame uniformly. Section 3 is devoted to repairing this.

3 Almost sure models via a poset encoding

3.1 The Kleitman–Rothschild theorem

The *height* of an element w of a finite poset is the largest size of a chain whose greatest element is w, and the height of the poset is the largest height of its elements. For a poset Q we write $L_i(Q)$ for the set of elements of height i, the ith *level* of Q; thus $L_1(Q)$ is the set $\min(Q)$ of minimal elements. We say that *almost every* poset has a given property if the proportion of posets on an m-element set that have the property tends to 1 as $m \to \infty$. Throughout, P_m denotes the number of posets on an m-element set.

Let Y be a finite set and let $\pi = (X_1, X_2, X_3)$ be an ordered partition of Y into three sets, some of which may be empty. Following Brightwell [4], we write $\mathscr{A}(\pi) = \mathscr{A}(X_1, X_2, X_3)$ for the set of partial orders $\le$ on Y such that

(L1) if $x \in X_i$, $y \in X_j$ and $x < y$, then $i < j$;

(L2) $x < z$ for all $x \in X_1$ and all $z \in X_3$.

An order in $\mathscr{A}(\pi)$ is determined by the set of pairs $(x, y) \in X_1 \times X_2$ with $x < y$ together with the set of pairs $(y, z) \in X_2 \times X_3$ with $y < z$. Conversely, every choice of these two sets of pairs yields an order in $\mathscr{A}(\pi)$, transitivity being guaranteed by (L2). Hence $|\mathscr{A}(\pi)| = 2^{|X_1||X_2|+|X_2||X_3|}$, and under the uniform probability measure $\Pr_\pi$ on $\mathscr{A}(\pi)$ the events $x < y$, for $(x, y) \in X_1 \times X_2$, and $y < z$, for $(y, z) \in X_2 \times X_3$, are mutually independent and each has probability $\frac{1}{2}$.

Call a partition $\pi = (X_1, X_2, X_3)$ of an m-element set *balanced* if

$$\big||X_2| - \tfrac{m}{2}\big| < \log m \qquad \text{and} \qquad \big||X_i| - \tfrac{m}{4}\big| < \sqrt{m}\,\log m \quad (i = 1, 3).$$

Theorem 3.1 (Kleitman and Rothschild [15]; see [4])**.** *Almost every partial order on an m-element set Y lies in $\mathscr{A}(\pi)$ for some balanced partition π of Y. Moreover $\log_2 P_m = m^2/4 + 3m/2 + O(\log m)$.*

The next lemma is the form in which we use Theorem 3.1.

Lemma 3.2. *For each m let $\mathcal{B}_m$ be a set of partial orders on an m-element set Y, and suppose that $\max_\pi \Pr_\pi[\mathcal{B}_m] \to 0$ as $m \to \infty$, the maximum being over the balanced partitions π of Y. Then $|\mathcal{B}_m|/P_m \to 0$.*

Proof. Let m be large enough that every balanced partition (X_1, X_2, X_3) has $|X_1|, |X_3| \ge m/5$ and $|X_2| \ge m/3$, and that the quantity δ_m defined below is less than 1. For a balanced partition $\pi = (X_1, X_2, X_3)$ let $\mathscr{A}^\circ(\pi)$ be the set of orders in $\mathscr{A}(\pi)$ in which every element of X_2 lies above some element of X_1 and every element of X_3 lies above some element of X_2. By (L1), in such an order the elements of X_1, X_2 and X_3 are exactly the elements of height 1, 2 and 3 respectively. So π is determined by the order, and the sets $\mathscr{A}^\circ(\pi)$, for distinct balanced π, are pairwise disjoint. By independence,

$$\Pr_\pi\big[\mathscr{A}(\pi) \setminus \mathscr{A}^\circ(\pi)\big] \;\le\; |X_2|\,2^{-|X_1|} + |X_3|\,2^{-|X_2|} \;\le\; m\,2^{-m/5} + m\,2^{-m/3} \;=:\; \delta_m,$$

and therefore

$$\sum_\pi |\mathscr{A}(\pi)| \;\le\; (1-\delta_m)^{-1} \sum_\pi |\mathscr{A}^\circ(\pi)| \;\le\; (1-\delta_m)^{-1} P_m,$$

the sums being over balanced partitions. Let $\mathcal{E}_m$ be the set of partial orders on Y that lie in no $\mathscr{A}(\pi)$ with π balanced. Then

$$|\mathcal{B}_m| \;\le\; |\mathcal{E}_m| + \sum_\pi |\mathcal{B}_m \cap \mathscr{A}(\pi)| \;\le\; |\mathcal{E}_m| + \max_\pi \Pr_\pi[\mathcal{B}_m] \cdot (1-\delta_m)^{-1} P_m,$$

and both terms are $o(P_m)$, the first by Theorem 3.1. □

We now isolate the properties of a typical poset that will be needed. Unlike "$|L_1| = m/4 + o(m)$", each of them is a property of an individual poset.

Definition 3.3. *A finite poset Q is* k-good *if*

(G1) *Q has height 3, and $|L_1(Q)| \geq k+1$;*

(G2) *$x < z$ for all $x \in L_1(Q)$ and all $z \in L_3(Q)$;*

(G3) *every element of $L_2(Q)$ lies above at least $k+1$ elements of $L_1(Q)$ and below at least one element of $L_3(Q)$;*

(G4) *for all pairwise distinct $x, b_1, \ldots, b_k \in L_1(Q)$ and every $S \subseteq K$ there is $y \in L_2(Q)$ such that $x < y$ and, for each $i \in K$, $b_i < y$ if and only if $i \in S$.*

Lemma 3.4. *Almost every poset Q on m points is k-good and satisfies $\big||L_1(Q)| - m/4\big| < \sqrt{m}\,\log m$.*

Proof. Let $\pi = (X_1, X_2, X_3)$ be a balanced partition of an m-element set, with m large enough that $|X_1|, |X_3| \geq m/5 \geq k+1$ and $|X_2| \geq m/3$, and consider a random order drawn from $\mathscr{A}(\pi)$ according to $\Pr_\pi$. By the independence of the relations between consecutive layers, and using $|X_i| \leq m$:

(a) some $y \in X_2$ lies above at most k elements of X_1 with probability at most

$$|X_2| \sum_{j \leq k} \binom{|X_1|}{j} 2^{-|X_1|} \;\leq\; m(m+1)^k 2^{-m/5};$$

(b) the probability that some $y \in X_2$ lies below no element of X_3 is at most $|X_2|\,2^{-|X_3|} \leq m\,2^{-m/5}$;

(c) the probability that some $z \in X_3$ lies above no element of X_2 is at most $|X_3|\,2^{-|X_2|} \leq m\,2^{-m/3}$;

(d) for fixed pairwise distinct $x, b_1, \ldots, b_k \in X_1$ and fixed $S \subseteq K$, each $y \in X_2$ satisfies "$x < y$, and $b_i < y$ if and only if $i \in S$" with probability $2^{-(k+1)}$, independently for different y; so the probability that for some choice of $x, b_1, \ldots, b_k$ and S no such y exists is at most $m^{k+1} 2^k \big(1 - 2^{-(k+1)}\big)^{m/3}$.

All four bounds tend to 0 as $m \to \infty$, and they do not depend on π. Suppose none of the four exceptional events occurs. By (a), (c) and (L1), the elements of X_1, X_2, X_3 are exactly those of height 1, 2, 3, so that $L_i(Q) = X_i$ and, as $X_3 \neq \emptyset$, the height is 3. Then (G1) holds because $|X_1| \geq k+1$, (G2) is (L2), (G3) is (a) together with (b), and (G4) is (d); moreover $\big||L_1(Q)| - m/4\big| = \big||X_1| - m/4\big| < \sqrt{m}\log m$. The lemma now follows from Lemma 3.2, applied to the set $\mathcal{B}_m$ of posets that fail to be k-good or violate the bound on $|L_1(Q)|$. □

3.2 The encoding

The obstacle to importing the Kleitman–Rothschild theorem into the intuitionistic setting is that the uniform measure on $\mathrm{Mod}_n^{\mathsf{IPC}}$ is *not* the uniform measure on posets. For the modal logics K, T, $\mathsf{S4}$ and GL every frame on n points carries exactly 2^{kn} models, so a property holding in almost all frames holds in almost all models for trivial reasons. For IPC a frame F carries $U(F)^k$ models, as recorded in (1). Sampling a model

uniformly therefore samples a poset with probability proportional to $U(F)^k$, and Theorem 3.1 says nothing directly about that measure.

We avoid the weighted count altogether. The observation is that an up-set is the same thing as an extra minimal point, so that an n-element IPC-model over k variables *is* a poset on $n+k$ points in which k designated points are minimal. This replaces the weighted measure by the uniform measure on posets conditioned on an event whose probability is bounded away from zero, and conditioning on such an event preserves every almost sure property.

Put

$$X \;=\; \{1,\dots,n\}, \qquad A \;=\; \{a_1,\dots,a_k\}, \qquad Y \;=\; X \cup A, \qquad m \;=\; n+k,$$

with A disjoint from X. Let $\mathcal{P}_m$ be the set of partial orders on Y, so that $|\mathcal{P}_m| = P_m$, and let

$$\mathcal{G}_{n,k} \;=\; \big\{\, Q \in \mathcal{P}_m \;:\; a_i \text{ is a minimal element of } Q \text{ for each } i \in K \,\big\}.$$

We write a model $(X, \le, V)$ with underlying set X as the tuple $(F, U_1, \dots, U_k)$, where $F = (X, \le)$ and $U_i = V(p_i)$. For $Q \in \mathcal{P}_m$ we write $Q|_X$ for the restriction of Q to X and $\uparrow_Q(a)$ for the up-set of a in Q.

Lemma 3.5. *The map*

$$\Phi(Q) \;=\; \big(\, Q|_X,\; \uparrow_Q(a_1) \cap X,\; \dots,\; \uparrow_Q(a_k) \cap X \,\big)$$

is a bijection from $\mathcal{G}_{n,k}$ onto $\mathrm{Mod}_n^{\mathsf{IPC}}$. In particular

$$|\mathcal{G}_{n,k}| \;=\; |\,\mathrm{Mod}_n^{\mathsf{IPC}}\,| \;=\; \sum_F U(F)^k.$$

Proof. Write $U_i = \uparrow_Q(a_i) \cap X$. Since distinct a_i, a_j are both minimal they are incomparable, so $\uparrow_Q(a_i) = \{a_i\} \cup U_i$.

Φ *lands in* $\mathrm{Mod}_n^{\mathsf{IPC}}$. Each U_i is an up-set of $Q|_X$: if $x \in U_i$ and $x \le_Q y$ with $y \in X$, then $a_i \le_Q x \le_Q y$, so $y \in U_i$.

Injectivity. A partial order on Y is determined by its restriction to X, the pairs (a_i, x) with $x \in X$, the pairs (x, a_i) with $x \in X$, and the pairs (a_i, a_j) with $i \ne j$. For $Q \in \mathcal{G}_{n,k}$ the last two families are empty by minimality, the first is $Q|_X$ and the second is recorded by $U_1, \dots, U_k$.

Surjectivity. Given a model $(F, U_1, \dots, U_k)$ on X, let

$$\le_Q \;=\; \le_F \;\cup\; \{(a_i, a_i) : i \in K\} \;\cup\; \{(a_i, x) : i \in K,\; x \in U_i\}.$$

Reflexivity is clear. Antisymmetry holds because no element of X lies below any a_i and no a_i lies below a different a_j. For transitivity, the only configuration not internal to F is $a_i \le_Q x \le_Q y$ with $x, y \in X$, and then $y \in U_i$ because U_i is an up-set, so $a_i \le_Q y$. Each a_i is minimal in Q by construction, so $Q \in \mathcal{G}_{n,k}$, and $\Phi(Q) = (F, U_1, \dots, U_k)$. □

Remark 3.6. *As a check, take $n = 2$ and $k = 1$. There are three posets on two points, with 4, 3 and 3 up-sets respectively, so $|\,\mathrm{Mod}_2^{\mathsf{IPC}}\,| = 10$. Of the 19 posets on three labelled points the total number of minimal elements is 30, so by symmetry exactly 10 of them have a given point minimal.*

Lemma 3.7. $\displaystyle\lim_{n\to\infty} \frac{|\mathcal{G}_{n,k}|}{P_{n+k}} \;=\; 4^{-k}.$

Proof. Let Q be uniformly distributed on $\mathcal{P}_m$, and let ρ be a uniformly random permutation of Y, independent of Q. Relabelling by a fixed permutation is a bijection of $\mathcal{P}_m$ onto itself, so the relabelled order ρQ is again uniformly distributed on $\mathcal{P}_m$; and $\min(\rho Q) = \rho(\min Q)$. Conditionally on Q, the set $\rho(\min Q)$ is a uniformly random subset of Y of size $|\min Q|$, and a uniformly random ℓ-subset of the m-set Y contains the k-set A with probability $\binom{\ell}{k}/\binom{m}{k}$. Hence

$$\frac{|\mathcal{G}_{n,k}|}{P_m} \;=\; \Pr\bigl[A \subseteq \min(\rho Q)\bigr] \;=\; \mathbb{E}\left[\binom{|\min Q|}{k}\Big/\binom{m}{k}\right].$$

The quantity inside the expectation lies between 0 and 1. By Lemma 3.4, with probability tending to 1 we have $\bigl||\min Q| - m/4\bigr| < \sqrt{m}\log m$, and on this event the quantity inside the expectation is $4^{-k} + o(1)$, since k is fixed. Therefore the expectation tends to 4^{-k}. □

Corollary 3.8. $|\operatorname{Mod}_n^{\mathsf{IPC}}| \sim 4^{-k} P_{n+k}$, *and consequently*

$$\log_2 |\operatorname{Mod}_n^{\mathsf{IPC}}| \;=\; \frac{n^2}{4} + \frac{(k+3)\,n}{2} + O_k(\log n).$$

Proof. The first statement is Lemmas 3.5 and 3.7. The second follows from the estimate of P_m in Theorem 3.1, since $(n+k)^2/4 + 3(n+k)/2 = n^2/4 + (k+3)n/2 + O_k(1)$. □

So the number of IPC-models is $P_n \cdot 2^{kn/2 + O_k(\log n)}$: the average frame behaves as though it carried $2^{n/2}$ up-sets.

Corollary 3.9 (Transfer)**.** *Let $\mathcal{X}_m \subseteq \mathcal{P}_m$ satisfy $|\mathcal{X}_m|/P_m \to 0$. Then*

$$\lim_{n\to\infty} \frac{|\mathcal{X}_{n+k} \cap \mathcal{G}_{n,k}|}{|\mathcal{G}_{n,k}|} \;=\; 0.$$

Consequently, under the bijection Φ, every property holding in almost all finite posets holds in almost all finite IPC*-models.*

Proof. With $m = n + k$,

$$\frac{|\mathcal{X}_m \cap \mathcal{G}_{n,k}|}{|\mathcal{G}_{n,k}|} \;\le\; \frac{|\mathcal{X}_m|/P_m}{|\mathcal{G}_{n,k}|/P_m},$$

where the numerator tends to 0 by hypothesis and the denominator tends to $4^{-k} > 0$ by Lemma 3.7. □

3.3 The structure of almost all models

Proposition 3.10. *Let $\mathfrak{t}_n \subseteq \operatorname{Mod}_n^{\mathsf{IPC}}$ be the image under Φ of the set of k-good members of $\mathcal{G}_{n,k}$. Then*

$$\lim_{n\to\infty} \frac{|\mathfrak{t}_n|}{|\operatorname{Mod}_n^{\mathsf{IPC}}|} \;=\; 1.$$

Moreover, let $(F, U_1, \dots, U_k) = \Phi(Q) \in \mathfrak{t}_n$ with Q a k-good member of $\mathcal{G}_{n,k}$, and put $l_0 = L_1(Q) \setminus A$, $l_1 = L_2(Q)$ and $l_2 = L_3(Q)$. Then:

1. *F has height 3, its levels are $L_1(F) = l_0$, $L_2(F) = l_1$ and $L_3(F) = l_2$, and all three are non-empty;*
2. *$U_i \cap l_0 = \varnothing$ for each $i \in K$;*

3. $l_2 \subseteq U_i$ *for each* $i \in K$*;*

4. $x < z$ *for all* $x \in l_0$ *and* $z \in l_2$*, and every element of* l_1 *lies below some element of* l_2*;*

5. *for every* $x \in l_0$ *and every* $S \subseteq K$ *there is* $y \in l_1$ *with* $x < y$ *and* $\{\, i \in K : y \in U_i \,\} = S$.

Proof. By Lemma 3.4 the posets on m points that are not k-good form a vanishing proportion of $\mathcal{P}_m$, so the density statement follows from Corollary 3.9 and the bijectivity of Φ.

Fix Q as in the statement and write $L_j = L_j(Q)$. Since each a_i is minimal in Q we have $A \subseteq L_1$, and so $X = l_0 \cup l_1 \cup l_2$ by (G1).

(1) The set l_0 is non-empty because $|L_1| \geq k+1$, and its elements are minimal in F. Let $y \in L_2$. Every element strictly below y in Q lies in L_1, and by (G3) at least one of them lies outside A; so y has height 2 in F. Let $z \in L_3$. There are $y \in L_2$ with $y < z$ and, by what was just shown, $x \in l_0$ with $x < y$; so z has height at least 3 in F, and hence exactly 3. Finally $l_2 = L_3 \neq \emptyset$ because Q has height 3, and then $l_1 \neq \emptyset$.

(2) The elements of L_1 are minimal in Q, so none of them lies strictly above a_i; hence $U_i \cap l_0 = \varnothing$.

(3) By (G2) applied to $a_i \in L_1$, every element of L_3 lies above a_i.

(4) This is (G2) together with the second half of (G3).

(5) Apply (G4) to $x, a_1, \ldots, a_k$, which are pairwise distinct elements of L_1 because $x \in L_1 \setminus A$. It provides $y \in L_2 = l_1$ with $x < y$ and with $a_i < y$ if and only if $i \in S$; and $a_i < y$ means precisely that $y \in U_i$. □

We refer to the members of $\mathfrak{t}_n$ as *typical* models. Proposition 3.10 says that almost every model has a three-level frame carrying a valuation that is empty on the bottom level and full on the top level, and in which every bottom point sees middle points of every possible valuation. No count of up-sets is needed at any point.

4 The representative model and the transfer theorem

We now define a single finite model $\mathcal{A}_k$, of size independent of n, and show that every typical model is bisimilar to it. Combined with Corollary 2.6, this reduces almost sure validity to validity in $\mathcal{A}_k$.

Definition 4.1. *The* representative model $\mathcal{A}_k$ *has underlying set*

$$\{\bot\} \;\cup\; \{m_B : B \subseteq K\} \;\cup\; \{\top\},$$

ordered by $\bot < m_B < \top$ *for every* $B \subseteq K$*, with no other strict relations, and has valuation* $V(p_i) = \{m_B : i \in B\} \cup \{\top\}$.

Each $V(p_i)$ is an up-set, so $\mathcal{A}_k$ is a model; in the notation of Section 2, $V(\bot) = \emptyset$, $V(m_B) = B$ and $V(\top) = K$. Its middle layer has 2^k points, one for each valuation. The underlying order is shown in Figure 1.

It will be convenient to have a slightly smaller model to hand. The point m_K satisfies every variable and its only strict successor is $\top$. Let $\mathcal{B}_k$ be the submodel of $\mathcal{A}_k$ obtained by deleting m_K.

Lemma 4.2. *The map* $b\colon \mathcal{A}_k \to \mathcal{B}_k$ *given by* $b(m_K) = \top$ *and* $b(z) = z$ *otherwise is a surjective p-morphism. Hence* $\mathcal{A}_k \models \varphi$ *if and only if* $\mathcal{B}_k \models \varphi$*, for every* $\varphi \in \mathcal{L}_k$.

Proof. Conditions (i)–(iii) of Definition 2.7 are immediate from Definition 4.1; the rest is Corollary 2.6. □

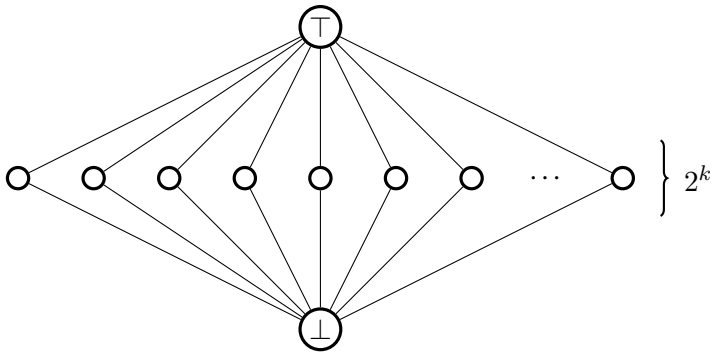


Figure 1: The Hasse diagram of $\mathcal{A}_k$: a bottom point, the 2^k middle points m_B, and a top point.

Remark 4.3. *For $k = 1$ the model $\mathcal{B}_1$ is the three-element chain $\bot < m_\emptyset < \top$, in which p_1 holds only at $\top$. It admits a further collapse: sending both $\bot$ and $m_\emptyset$ to the bottom point is a surjective p-morphism onto the two-element chain $\mathbf{C}_2$ of Example 2.11.*

Theorem 4.4. *Let $\mathfrak{M} = (X, \leq, V) \in \mathfrak{t}_n$ be a typical model, with levels l_0, l_1, l_2 as in Proposition 3.10. Define $\beta \colon X \to \mathcal{A}_k$ by*

$$\beta(x) = \bot \ \ (x \in l_0), \qquad \beta(y) = m_{V(y)} \ \ (y \in l_1), \qquad \beta(z) = \top \ \ (z \in l_2).$$

Then the graph of β is a total surjective bisimulation between $\mathfrak{M}$ and $\mathcal{A}_k$. Consequently $\mathfrak{M} \models \varphi$ if and only if $\mathcal{A}_k \models \varphi$, for every $\varphi \in \mathcal{L}_k$.

Proof. We use clauses (1)–(5) of Proposition 3.10, recalling that $V(p_i)$ is the set U_i of that proposition.

(atoms). By (2) no variable holds at a point of l_0, and none holds at $\bot$. By (3) every variable holds at each point of l_2, and every variable holds at $\top$. For $y \in l_1$ the points y and $m_{V(y)}$ satisfy the same variables by Definition 4.1.

(forth). Let $w \leq v$ in $\mathfrak{M}$; we show $\beta(w) \leq \beta(v)$. This is clear if $w = v$, and also if $w \in l_0$, since $\bot$ is the least element of $\mathcal{A}_k$. Otherwise $w < v$ and $w \in l_1$, because the points of l_2 are maximal by (1). Then v has height 3, so $v \in l_2$ by (1), and $\beta(v) = \top \geq \beta(w)$.

(back). Let $\beta(w) \leq v'$ in $\mathcal{A}_k$; we find $v \geq w$ with $\beta(v) = v'$. If $v' = \beta(w)$ we take $v = w$. Otherwise either $w \in l_0$ and $v' \in \{m_S : S \subseteq K\} \cup \{\top\}$, or $w \in l_1$ and $v' = \top$. If $w \in l_0$ and $v' = m_S$, clause (5) provides $y \in l_1$ with $w < y$ and $V(y) = S$, so that $\beta(y) = m_S$. If $v' = \top$, clause (4) provides $z \in l_2$ with $w < z$: any element of l_2 if $w \in l_0$, and a suitable one if $w \in l_1$.

The graph of β is total because β is a map defined on all of X. It is surjective: $\bot$ and $\top$ are values of β because l_0 and l_2 are non-empty by (1), and each m_S is a value of β by (5) applied to any $x \in l_0$. The final assertion is Corollary 2.6. □

5 The zero-one law for model validity

The zero-one law now follows in two short steps. The first is a general remark about densities.

Lemma 5.1. *Let $\mathfrak{t}_n \subseteq \mathrm{Mod}_n^{\mathsf{IPC}}$ with $|\mathfrak{t}_n|/|\mathrm{Mod}_n^{\mathsf{IPC}}| \to 1$, and let $X_n \subseteq \mathrm{Mod}_n^{\mathsf{IPC}}$. Then $|X_n|/|\mathrm{Mod}_n^{\mathsf{IPC}}| \to 1$ if and only if $|X_n \cap \mathfrak{t}_n|/|\mathfrak{t}_n| \to 1$, and $|X_n|/|\mathrm{Mod}_n^{\mathsf{IPC}}| \to 0$ if and only if $|X_n \cap \mathfrak{t}_n|/|\mathfrak{t}_n| \to 0$.*

Proof. Write $N_n = |\operatorname{Mod}_n^{\mathsf{IPC}}|$ and $e_n = N_n - |\mathfrak{t}_n|$, so $e_n/N_n \to 0$. From $|X_n \cap \mathfrak{t}_n| \le |X_n| \le |X_n \cap \mathfrak{t}_n| + e_n$ and $|\mathfrak{t}_n|/N_n \to 1$ it follows that $|X_n|/N_n - |X_n \cap \mathfrak{t}_n|/|\mathfrak{t}_n| \to 0$, which gives both claims. □

Theorem 5.2. *Let $\varphi \in \mathcal{L}_k$. If $\mathcal{A}_k \models \varphi$ then φ is almost surely valid; otherwise φ is almost surely invalid. In particular*

$$\lim_{n\to\infty} \frac{|\operatorname{Mod}_n^{\mathsf{IPC}}(\varphi)|}{|\operatorname{Mod}_n^{\mathsf{IPC}}|} \in \{0,1\}$$

for every $\varphi \in \mathcal{L}_k$, and the zero-one law holds.

Proof. By Theorem 4.4, every $\mathfrak{M} \in \mathfrak{t}_n$ satisfies $\mathfrak{M} \models \varphi$ if and only if $\mathcal{A}_k \models \varphi$. So $\operatorname{Mod}_n^{\mathsf{IPC}}(\varphi) \cap \mathfrak{t}_n$ is either all of $\mathfrak{t}_n$ or empty, according as $\mathcal{A}_k \models \varphi$ or not. By Proposition 3.10 the class $\mathfrak{t}_n$ has density 1, so Lemma 5.1 applies with $X_n = \operatorname{Mod}_n^{\mathsf{IPC}}(\varphi)$ and gives the claim. □

Corollary 5.3. *A formula $\varphi \in \mathcal{L}_k$ is almost surely valid if and only if $\mathcal{B}_k \models \varphi$.*

Proof. Theorem 5.2 and Lemma 4.2. □

Remark 5.4. *The measure on $\operatorname{Mod}_n^{\mathsf{IPC}}$ depends on k, and the measures for different values of k are not marginals of one another: if a model over $p_1, \dots, p_{k+1}$ is chosen uniformly and the variable p_{k+1} is then forgotten, the resulting model over $p_1, \dots, p_k$ has a given frame F with probability proportional to $U(F)^{k+1}$ rather than $U(F)^k$. Nevertheless, whether a formula is almost surely valid does not depend on the ambient number of variables. Indeed, let $j \le k$ and $\varphi \in \mathcal{L}_j$, and regard $\mathcal{A}_k$ as a model for the variables $p_1, \dots, p_j$ only. The map $\mathcal{A}_k \to \mathcal{A}_j$ that sends $\bot$ to $\bot$, $\top$ to $\top$ and m_B to $m_{B \cap \{1,\dots,j\}}$ is then a surjective p-morphism, so $\mathcal{A}_k \models \varphi$ if and only if $\mathcal{A}_j \models \varphi$. By Theorem 5.2, φ is almost surely valid with respect to models over k variables if and only if it is so with respect to models over j variables.*

6 An axiomatisation of the almost surely valid formulae

We show that the set of almost surely valid formulae has an axiomatisation resembling in spirit the theory of the random graph. Namely, for each $k \ge 1$ we exhibit a formula $\sigma_k \in \mathcal{L}_k$ such that for every $\varphi \in \mathcal{L}_k$,

$$\varphi \text{ is almost surely valid} \iff \vdash_{\mathsf{IPC}} \sigma_k \to \varphi. \tag{$\dagger$}$$

We make use of certain formulae, which we discovered by careful analysis of the frame corresponding to the free k-generated Heyting algebra, described first by Bellissima [2]. Once we have climbed this ladder we can discard it, so neither Heyting algebras nor Bellissima's construction will make explicit appearance.

Throughout this section, for a model $(W, \le, V)$ and $w \in W$ we write $w \models_V \varphi$ instead of $(W, \le, V), w \models \varphi$; the letters A and B range over *proper* subsets of $K = \{1, \dots, k\}$, and $\subsetneq$ denotes proper inclusion; and we adopt the conventions $\bigwedge \emptyset = \top$ and $\bigvee \emptyset = \bot$. Recall that $V(w) = \{i \in K : w \models_V p_i\}$.

6.1 Reductions

Definition 6.1. *Let $(W, \le, V)$ be a model and let $w, v \in W$. Let $\sim = \{(u,u) : u \in W\} \cup \{(w,v),(v,w)\}$, so that $\sim$ is the smallest equivalence relation on W identifying w and v. We call $\sim$ the (w,v)-*collapse*. Let W' be $W/{\sim}$, let $\le'$ be defined on W' by $[a]_\sim \le' [b]_\sim$ if and only if $x \le y$ holds for some $x \in [a]_\sim$ and some $y \in [b]_\sim$, and let $V'(p_i) = \{[a]_\sim : a \in V(p_i)\}$.*

1. *If* $\uparrow w \setminus \{w\} = \uparrow v$ *and* $V(w) = V(v)$*, then the natural quotient map* $f_\sim \colon W \to W'$ *is called an* α-reduction*, and* $(W', \leq', V')$ *an* α-reduct *of* $(W, \leq, V)$*.*

2. *If* $w \neq v$*,* $\uparrow w \setminus \{w\} = \uparrow v \setminus \{v\}$ *and* $V(w) = V(v)$*, then the natural quotient map* $f_\sim \colon W \to W'$ *is called a* β-reduction*, and* $(W', \leq', V')$ *a* β-reduct *of* $(W, \leq, V)$*.*

Lemma 6.2. *Let* $(W, \leq, V)$ *be a model, and let* $\gamma \in \{\alpha, \beta\}$*. Any* γ*-reduct* $(W', \leq', V')$ *is a model, and the quotient map* $f_\sim \colon W \to W'$ *is a surjective p-morphism. In particular a model and its* γ*-reducts are bisimilar and validate the same formulae.*

Proof. The fact that $(W', \leq')$ is a poset and that $f_\sim$ is a p-morphism of frames was shown by Bezhanishvili [3]; it is also easily verified directly. The assumption $V(w) = V(v)$ implies that V' is a well-defined valuation satisfying condition (iii) of Definition 2.7. □

Let us call a model *reduced* if it is rooted and admits no α- or β-reduction. Every finite rooted model is bisimilar to a reduced model: apply reductions for as long as possible; each reduct is rooted, has fewer points than its predecessor, and is bisimilar to it by Lemma 6.2.

6.2 The axioms σ_k

Definition 6.3. *For any* $A \subsetneq K$*, define*

$$\mu_A = \bigwedge_{i \in K} \neg\neg p_i \wedge \bigwedge_{j \in A} p_j \wedge \bigwedge \{p_r \leftrightarrow p_s : r, s \in K \setminus A\},$$
$$\tau_A = \bigvee \{\mu_B : B \subsetneq K,\ B \neq A\},$$

and put

$$\tau = \bigvee \{\mu_B : B \subsetneq K\}, \qquad \sigma_k = \bigwedge_{i \in K} \neg\neg p_i \wedge \bigwedge \{(\mu_B \to \tau_B) \to \tau : B \subsetneq K\}.$$

Remark 6.4. *For* $k = 1$ *the only proper subset of* K *is* $\emptyset$*; the formula* $\mu_\emptyset$ *is equivalent to* $\neg\neg p_1$*, while* $\tau_\emptyset = \bot$ *and* $\tau = \mu_\emptyset$*. So* σ_1 *is equivalent over* IPC *to* $\neg\neg p_1 \wedge (\neg\neg\neg p_1 \to \neg\neg p_1)$*, and hence to* $\neg\neg p_1$*, the second conjunct being a consequence of the first.*

We now analyse the finite models of σ_k, by which we mean the models in which σ_k is valid. Let $(W, \leq, V)$ be such a model. Then $w \models_V \neg\neg p_i$ for every $w \in W$ and $i \in K$, so above every point there is a point at which p_i holds; in particular every maximal element of W satisfies every variable. Any two distinct maximal elements therefore have the same valuation K and the same (empty) strict up-set, and their collapse is a β-reduction. Consequently a reduced model of σ_k has exactly one maximal element, which is then its greatest element. We denote it by $\top$, and we denote the root by $\bot$. Thus $V(\top) = K$, and $\top \models_V \mu_A$ for every $A \subsetneq K$. Note also that every $w \neq \top$ lies below some subcover of $\top$.

Lemma 6.5. *Let* $(W, \leq, V)$ *be a reduced model of* σ_k*, and let* $w \in W$ *be a subcover of* $\top$*. Then* $w \not\models_V p_i$ *for some* $i \in K$*.*

Proof. Suppose $w \models_V p_i$ for all $i \in K$. Then $V(w) = V(\top)$ and $\uparrow w \setminus \{w\} = \{\top\} = \uparrow \top$, so the $(w, \top)$-collapse gives an α-reduction of $(W, \leq, V)$, contradicting the assumption that $(W, \leq, V)$ is reduced. □

Lemma 6.6. *Let $(W, \leq, V)$ be a reduced model of σ_k, let $w \in W \setminus \{\top\}$, and let $A \subsetneq K$. Assume $w \models_V \mu_A$. Then the following hold:*

1. *For all $r, s \notin A$ the variables p_r and p_s cannot be distinguished in the up-set $\uparrow w$.*
2. *For each $s \notin A$, we have $w \not\models_V p_s$; hence $V(w) = A$.*
3. *For any $B \subsetneq K$ with $B \neq A$, we have $w \not\models_V \mu_B$.*
4. *We have $w \not\models_V \tau_A$, and therefore $w \not\models_V \mu_A \to \tau_A$.*

Proof. To prove (1) it suffices to note that $w \models_V p_r \leftrightarrow p_s$ if and only if for all $u \geq w$ we have $u \models_V p_r \iff u \models_V p_s$.

For (2), let w' be a subcover of $\top$ with $w \leq w'$. By Lemma 6.5 there is $i \in K$ with $w' \not\models_V p_i$, hence $w \not\models_V p_i$ by persistence, and $i \notin A$ because $w \models_V p_j$ for all $j \in A$. For $s \notin A$ the conjunct $p_s \leftrightarrow p_i$ of μ_A now gives $w \not\models_V p_s$.

To prove (3) suppose otherwise, that is, $w \models_V \mu_B$ for some $B \neq A$. Then $B \subseteq V(w) = A$, so $B \subsetneq A$. Take $t \in A \setminus B$ and $s \in K \setminus A$; then $t, s \notin B$, while $w \models_V p_t$ and $w \not\models_V p_s$. This contradicts the conjunct $p_t \leftrightarrow p_s$ of μ_B. Then (4) follows from (3) by definition of τ_A. □

Lemma 6.7. *Let $(W, \leq, V)$ be a reduced model of σ_k, and let $w, u \in W \setminus \{\top\}$. Assume $w \models_V \mu_A$ and $u \models_V \mu_B$ with $A \neq B$. Then w and u are incomparable.*

Proof. Suppose $w \leq u$. By persistence $u \models_V \mu_A$, which together with $u \models_V \mu_B$ contradicts Lemma 6.6(3). The case $u \leq w$ is symmetric. □

Lemma 6.8. *Let $(W, \leq, V)$ be a reduced model of σ_k, and let $w, u \in W$ be subcovers of $\top$. Then the following hold:*

1. *If $w \neq u$ then $V(w) \neq V(u)$.*
2. *There is a unique $A \subsetneq K$ such that $w \models_V \mu_A$, namely $A = V(w)$.*

Proof. For (1), note that $\uparrow w \setminus \{w\} = \{\top\} = \uparrow u \setminus \{u\}$; so if $w \neq u$ and $V(w) = V(u)$, the (w, u)-collapse would be a β-reduction. For (2), let $A = V(w)$, a proper subset of K by Lemma 6.5. Clearly $w \models_V \bigwedge_{j \in A} p_j$, and $w \models_V \bigwedge_{i \in K} \neg\neg p_i$ because σ_k is valid. Moreover, since $\uparrow w = \{w, \top\}$ and $\top \models_V p_i$ for every $i \in K$, we get $w \models_V \bigwedge\{p_r \leftrightarrow p_s : r, s \in K \setminus A\}$, proving $w \models_V \mu_A$. Uniqueness of A follows by Lemma 6.6(3). □

Corollary 6.9. *Let $(W, \leq, V)$ be a reduced model of σ_k, let $w \in W$ be a subcover of $\top$, and let $A \subsetneq K$. Then $w \models_V \mu_A$ if and only if both $w \models_V \bigwedge_{i \in A} p_i$ and $w \not\models_V \bigvee_{j \notin A} p_j$.*

Before we proceed further, it will be useful to take care of reduced models of σ_k that are "too small to be interesting".

Lemma 6.10. *Let $(W, \leq, V)$ be a reduced model of σ_k. The following hold:*

1. *If $|W| = 1$, then $W = \{\top\}$ and $V(\top) = K$.*
2. *If $|W| = 2$, then $W = \{\bot, \top\}$ with $\bot < \top$, $V(\top) = K$ and $V(\bot) \subsetneq K$.*
3. *If $k = 1$, then $|W| \leq 2$.*

Proof. Point (1) is clear. For (2), a rooted model with two elements is a chain $\bot < \top$, in which $\bot$ is a subcover of $\top$; apply Lemma 6.5.

For (3), let $k = 1$ and suppose $|W| \geq 2$. Every $w \neq \top$ lies below a subcover of $\top$, at which p_1 fails by Lemma 6.5; so $V(w) = \emptyset$ for every $w \neq \top$. By Lemma 6.8(1), $\top$ then has a unique subcover c, and every $w \neq \top$ satisfies $w \leq c$. If there were an element below c other than c, we could choose such an element w maximal; then $\uparrow w \setminus \{w\} = \{c, \top\} = \uparrow c$ and $V(w) = V(c)$, so the (w, c)-collapse would be an α-reduction. Hence $W = \{c, \top\}$. □

Lemma 6.11. *Let $(W, \leq, V)$ be a reduced model of σ_k, and let w_1 and w_2 be distinct subcovers of $\top$. Then $u \not\models_V \tau$ for any $u \in W$ with $u \leq w_1$ and $u \leq w_2$.*

Proof. Let $N_{w_1} = \{A \subsetneq K : w_1 \not\models_V \mu_A\}$ and $N_{w_2} = \{A \subsetneq K : w_2 \not\models_V \mu_A\}$. By Lemma 6.8, w_1 and w_2 satisfy μ_A for exactly one A each, and not for the same one; so $N_{w_1} \cup N_{w_2}$ is the set of all proper subsets of K. Since $u \leq w_1$, for each $A \in N_{w_1}$ we have $u \not\models_V \mu_A$ by persistence; since $u \leq w_2$, for each $B \in N_{w_2}$ we have $u \not\models_V \mu_B$. Hence $u \not\models_V \mu_A$ for every $A \subsetneq K$, and so $u \not\models_V \tau$. □

Lemma 6.12. *Let $(W, \leq, V)$ be a reduced model of σ_k. Then there are at most $2^k - 1$ subcovers of $\top$ in W.*

Proof. By Lemma 6.5, $V(w)$ is a proper subset of K for each subcover w of $\top$. By Lemma 6.8(1), these subsets are all distinct. □

Lemma 6.13. *Let $(W, \leq, V)$ be a reduced model of σ_k. If $k \geq 2$ and there are at least two subcovers of $\top$ in W, then there are at least $2^k - 1$ subcovers of $\top$ in W.*

Proof. Suppose there are j subcovers of $\top$ with $2 \leq j < 2^k - 1$, call them $w_1, \ldots, w_j$. By Lemma 6.8(2) there is some $A \subsetneq K$ such that $w_l \not\models_V \mu_A$ for every $l \leq j$. Every $u < \top$ lies below some w_l, hence $u \not\models_V \mu_A$ for every $u < \top$, by persistence. Moreover, $\top \models_V \mu_A$, and $\top \models_V \tau_A$ because $k \geq 2$. It follows that $u \models_V \mu_A \to \tau_A$ for every $u \in W$; in particular, $\bot \models_V \mu_A \to \tau_A$. On the other hand, by Lemma 6.11, we have $\bot \not\models_V \tau$. Hence, $\bot \not\models_V (\mu_A \to \tau_A) \to \tau$, contradicting the assumption that $(W, \leq, V)$ is a model of σ_k. □

Corollary 6.14. *Let $(W, \leq, V)$ be a reduced model of σ_k. If $k \geq 2$ and there are at least two subcovers of $\top$ in W, then there are precisely $2^k - 1$ of them, and $w \mapsto V(w)$ is a bijection from the set of subcovers of $\top$ onto the set of proper subsets of K.*

Lemma 6.15. *Let $(W, \leq, V)$ be a reduced model of σ_k. Let C be the set of subcovers of $\top$ in W and let $u \in W$. If $|\uparrow u \cap C| \geq 2$, then $C \subseteq \uparrow u$.*

Proof. Assume $|\uparrow u \cap C| \geq 2$. Then $|C| \geq 2$, and so $k \geq 2$ by Lemma 6.10(3). Applying Lemma 6.11 we obtain $u \not\models_V \tau$. To get a contradiction, suppose there exists $w \in C \setminus \uparrow u$. Let $A = V(w)$, so that $w \models_V \mu_A$. By Lemma 6.8, $w' \not\models_V \mu_A$ for each $w' \in \uparrow u \cap C$. Every $v \in \uparrow u \setminus \{\top\}$ lies below some $w' \in \uparrow u \cap C$, hence $v \not\models_V \mu_A$ for any such v. Since $\top \models_V \tau_A$, we get that $u \models_V \mu_A \to \tau_A$. It follows that $u \not\models_V (\mu_A \to \tau_A) \to \tau$, contradicting the assumption that $(W, \leq, V)$ is a model of σ_k. □

Corollary 6.16. *Let $(W, \leq, V)$ be a reduced model of σ_k. Let C be the set of subcovers of $\top$ in W. If $u \in W \setminus (C \cup \{\top\})$, then $C \subseteq \uparrow u$ or $|C \cap \uparrow u| = 1$.*

Lemma 6.17. *Let $(W, \leq, V)$ be a reduced model of σ_k with $k \geq 2$, and let C be the set of subcovers of $\top$ in W. Then there is no $u \in W \setminus (C \cup \{\top\})$ with $|C \cap \uparrow u| = 1$.*

Proof. Suppose u is such an element, and let $C \cap {\uparrow}u = \{w\}$. Then $u < w$, so we may choose z with $u \leq z \prec w$. Since ${\uparrow}z \subseteq {\uparrow}u$, the only subcover of $\top$ above z is w. Hence every $y > z$ other than $\top$ satisfies $y \leq w$, and therefore $y = w$ because $z \prec w$; that is, ${\uparrow}z = \{z, w, \top\}$.

Let $A = V(w)$, so that $w \models_V \mu_A$ by Lemma 6.8(2). By persistence $V(z) \subseteq A$. If $V(z) = A$, then, since ${\uparrow}z \setminus \{z\} = \{w, \top\} = {\uparrow}w$, the (z, w)-collapse would be an α-reduction, which is impossible in a reduced model. So $V(z) \subsetneq A$, whence $z \not\models_V \mu_A$. For $B \neq A$ we have $w \not\models_V \mu_B$ by Lemma 6.6(3), hence $z \not\models_V \mu_B$ by persistence. Therefore $z \not\models_V \tau$.

Now fix some $B \subsetneq K$ with $B \neq A$, which exists because $k \geq 2$. Among the points of ${\uparrow}z = \{z, w, \top\}$ only $\top$ satisfies μ_B, and $\top \models_V \tau_B$ because $\top \models_V \mu_A$. Hence $z \models_V \mu_B \to \tau_B$, and since $z \not\models_V \tau$ we get $z \not\models_V (\mu_B \to \tau_B) \to \tau$, contradicting the validity of σ_k. □

Lemma 6.18. *Let $(W, \leq, V)$ be a reduced model of σ_k with $k \geq 2$, and suppose that the set C of subcovers of $\top$ has at least two elements. Then $W = \{\bot\} \cup C \cup \{\top\}$, and $V(\bot) = \emptyset$.*

Proof. Let $D = W \setminus (C \cup \{\top\})$. The root $\bot$ belongs to D, for otherwise $W = {\uparrow}\bot \subseteq \{\bot, \top\}$ and $|C| \leq 1$.

Let $u \in D$. Since $u < \top$, some subcover of $\top$ lies above u; by Lemma 6.17 it is not the only one; so $C \subseteq {\uparrow}u$ by Lemma 6.15. By Corollary 6.14 some $c \in C$ has $V(c) = \emptyset$, and $u \leq c$ gives $V(u) = \emptyset$. Thus every element of D has empty valuation and lies below every element of C, and consequently

$$ {\uparrow}u \setminus \{u\} \;=\; \{v \in D : u < v\} \cup C \cup \{\top\} \qquad \text{for every } u \in D. \tag{2} $$

If u and u' were distinct maximal elements of D, then by (2) they would have the same strict up-set $C \cup \{\top\}$, and the same valuation, so that the (u, u')-collapse would be a β-reduction. Hence D has a unique maximal element u^*, which is therefore the greatest element of D. If $D \neq \{u^*\}$, let u be a maximal element of $D \setminus \{u^*\}$. Then $u < u^*$ and, by (2), ${\uparrow}u \setminus \{u\} = \{u^*\} \cup C \cup \{\top\} = {\uparrow}u^*$, while $V(u) = V(u^*)$; so the (u, u^*)-collapse would be an α-reduction. Hence $D = \{u^*\} = \{\bot\}$. □

Lemma 6.19. *Every reduced model of σ_k is isomorphic to a generated submodel of $\mathcal{B}_k$. Consequently every finite rooted model of σ_k is bisimilar to a generated submodel of $\mathcal{B}_k$.*

Proof. Let $(W, \leq, V)$ be a reduced model of σ_k, with greatest element $\top$, and let C be the set of subcovers of $\top$.

If $C = \emptyset$ then $W = \{\top\}$ and $V(\top) = K$, so the model is isomorphic to the generated submodel $\{\top\}$ of $\mathcal{B}_k$.

Suppose $|C| = 1$, say $C = \{c\}$. Then $W = \{c, \top\}$: for $k = 1$ this is Lemma 6.10(3), and for $k \geq 2$ any further element u would satisfy $|C \cap {\uparrow}u| = 1$, contradicting Lemma 6.17. By Lemma 6.5, $B = V(c)$ is a proper subset of K, and the model is isomorphic to the generated submodel $\{m_B, \top\}$ of $\mathcal{B}_k$.

Suppose finally that $|C| \geq 2$. Then $k \geq 2$ by Lemma 6.10(3). By Corollary 6.14, $c \mapsto V(c)$ is a bijection from C onto the set of proper subsets of K, and by Lemma 6.18, $W = \{\bot\} \cup C \cup \{\top\}$ with $V(\bot) = \emptyset$. The elements of C are pairwise incomparable, being subcovers of the same element, and $\bot < c < \top$ for each $c \in C$. Hence the map sending $\bot$ to $\bot$, each $c \in C$ to $m_{V(c)}$, and $\top$ to $\top$ is an isomorphism onto $\mathcal{B}_k$.

For the second statement, recall that a finite rooted model of σ_k is bisimilar to a reduced model, in which σ_k is again valid by Corollary 2.6. □

Lemma 6.20. *$\mathcal{B}_k \models \sigma_k$, for every $k \geq 1$.*

Proof. Every point of $\mathcal{B}_k$ lies below $\top$, and $V(\top) = K$. So each $\neg\neg p_i$ is valid in $\mathcal{B}_k$, and $\top \models_V \mu_B$ for every $B \subsetneq K$; in particular $\top \models_V \tau$. Next we claim that for proper subsets B and D of K,

$$m_D \models_V \mu_B \quad \text{if and only if} \quad B = D. \tag{3}$$

Recall that $\uparrow m_D = \{m_D, \top\}$ and $V(m_D) = D$. The variables p_r with $r \notin D$ all fail at m_D and all hold at $\top$, so every conjunct of μ_D holds at m_D. Conversely, if $m_D \models_V \mu_B$ then $B \subseteq D$; and if $B \neq D$ we may pick $t \in D \setminus B$ and $s \in K \setminus D$, so that $t, s \notin B$ while $m_D \models_V p_t$ and $m_D \not\models_V p_s$, contradicting the conjunct $p_t \leftrightarrow p_s$ of μ_B. This proves (3), from which it follows that $m_D \models_V \tau$ and $m_D \not\models_V \tau_D$ for every $D \subsetneq K$.

Now fix $B \subsetneq K$ and a point v of $\mathcal{B}_k$ such that $v \models_V \mu_B \to \tau_B$; we show that $v \models_V \tau$, which proves that $(\mu_B \to \tau_B) \to \tau$ holds at every point. Since $m_B \models_V \mu_B$ and $m_B \not\models_V \tau_B$, the formula $\mu_B \to \tau_B$ fails at m_B, and hence also at $\bot$. So v is either $\top$ or m_D for some $D \neq B$, and in both cases $v \models_V \tau$. □

Theorem 6.21. *Let $\varphi \in \mathcal{L}_k$. Then the following are equivalent:*

1. $\vdash_{\mathsf{IPC}} \sigma_k \to \varphi$;
2. *φ is almost surely valid.*

Proof. Suppose (1). By soundness, $\sigma_k \to \varphi$ is valid in $\mathcal{B}_k$, and σ_k is valid in $\mathcal{B}_k$ by Lemma 6.20. Hence $\mathcal{B}_k \models \varphi$, and φ is almost surely valid by Corollary 5.3.

Conversely, suppose that $\not\vdash_{\mathsf{IPC}} \sigma_k \to \varphi$. By the finite model property of IPC [5], there are a finite model and a point v of it at which σ_k holds and φ fails. The submodel generated by v is a finite rooted model in which σ_k is valid, by persistence, and φ is not. By Lemma 6.19 it is bisimilar to a generated submodel of $\mathcal{B}_k$, in which φ is therefore not valid, by Corollary 2.6. So $\mathcal{B}_k \not\models \varphi$, and φ is not almost surely valid, again by Corollary 5.3. □

7 Concluding remarks

We have shown that intuitionistic propositional logic obeys a zero-one law for model validity, and we have axiomatised the almost surely valid formulae. We close with three questions that our methods leave open.

Frame validity. Is every formula valid in almost all finite frames or in almost none? Here the relevant measure is simply the uniform measure on finite posets, to which Theorem 3.1 applies directly, so the difficulty addressed in Section 3 does not arise. But the encoding of that section gives no help either, since frame validity quantifies over all valuations on a given frame. For the provability logic GL the corresponding question was answered by Verbrugge [19].

Complexity. By Corollary 5.3, a formula $\varphi \in \mathcal{L}_k$ is almost surely valid if and only if it is true at the root of $\mathcal{B}_k$, a model with $2^k + 1$ points. So almost sure validity is decidable. We have not determined its exact complexity.

Infinitely many variables. Our measure is defined for a fixed finite number k of variables. By Remark 5.4 the notion of almost sure validity does not depend on k, but the measures for different k are not marginals of one another, and it is not clear what the appropriate probability measure on models over infinitely many variables should be.

Acknowledgments

The zero-one law and most of the results of this article were first obtained in Sigfrido Ciletti's Honours thesis, completed under the joint supervision of Marcel Jackson, Guillermo Badia and Tomasz Kowalski. The proof of Proposition 3.10 given here is new and replaces the counting argument of the thesis.

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